DEPARTMENT OF ELECTRONICS AND COMMUNICATION
BEC24502 - DIGITAL SIGNAL PROCESSING
Question Bank

MODULE-2

Explain the overlap add method of filtering long data sequences.

Explain the overlap save method with a neat block diagram.

Derive the decimation in time FFT algorithm for N = 8.

Compute the 8 point DFT of x(n) = {1, 2, 3, 4, 4, 3, 2, 1} using DIT FFT.

Compare DIT and DIF FFT algorithms.

Determine the number of complex multiplications saved by FFT over direct DFT computation for N = 64.

State and prove the symmetry property of the twiddle factor.

MODULE-3

Explain the design of FIR filters using the window method.

Compare rectangular, Hamming and Hanning windows.

Design a linear phase FIR low pass filter of length 7 with cutoff frequency 1 rad/sec using a Hamming window.

Explain the frequency sampling method of FIR filter design.

What are the advantages of FIR filters over IIR filters?

Realise the given FIR system in direct form I and direct form II.